

M2: Do High Follower Users Get More Engagement?

Target Database: data/data_vortex.db | Validation: Audited & Verified | Official Deliverable

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1. Analytical Objective & Hypothesis

The objective of challenge M2 is to rigorously test the common assumption that accounts with larger audience reach ($\geq 25,000$ followers) achieve higher engagement volume on their published posts than smaller accounts ($< 25,000$ followers). The SQL query segments creators into these two explicit tiers and compares post-level engagement metrics.

2. Relational Schema & Join Topology

This challenge requires a relational bridge between the creator dimension (users) and the interaction fact table (posts):

- **users:** Contains user_id (PK) and follower_count (INTEGER, range: 100 to 50,000).
- **posts:** Contains post_id (PK), user_id (FK), likes, shares, and comments.

An INNER JOIN on users.user_id = posts.user_id guarantees referential integrity without Cartesian artifacts.

3. Step-by-Step SQL Architecture

- Step 1 - Conditional Cohort Segmentation:** A conditional CASE WHEN u.follower_count ≥ 25000 THEN 'High Follower ($\geq 25,000$)' ELSE 'Low Follower ($< 25,000$)' END partitions the creator population exactly at the 25k boundary.
- Step 2 - Relational Inner Join:** Connects every post to its authoring creator's follower count. Database pre-validation confirmed exactly 0 orphan posts, guaranteeing 100% data retention (12,000 posts matched).
- Step 3 - Distinct Creator and Post Counting:** Utilizing COUNT(DISTINCT u.user_id) reveals an almost perfectly balanced cohort split: 742 High Follower creators (49.47%) and 758 Low Follower creators (50.53%).
- Step 4 - Aggregate Interaction Calculation:** Arithmetic means are computed for likes, shares, comments, and combined engagement (likes + shares + comments).
- Step 5 - Robustness Validation across Complete Cases:** Evaluated alongside complete-case filtering (AVG(likes + shares + comments)), confirming that both cohorts remain virtually identical.

4. Comparative Cohort Metrics Matrix

A granular comparison across both cohorts reveals negligible variation across all interaction dimensions:

Metric	High Follower ($\geq 25,000$)	Low Follower ($< 25,000$)	Absolute Delta	Relative Spread
Unique Creators	742 (49.5%)	758 (50.5%)	-16 users	-1.07%
Total Posts Published	5,925 (49.4%)	6,075 (50.6%)	-150 posts	-1.25%
Avg Posts per User	7.99	8.01	-0.02	-0.25%
Average Likes	2,496.30	2,487.62	+8.68	+0.35%
Average Shares	1,001.24	1,012.95	-11.71	-1.16%
Average Comments	504.20	504.49	-0.29	-0.06%
Avg Total Engagement (Official)	3,604.43	3,648.40	-43.97	-1.21%
Avg Total (Complete Cases)	4,000.91	4,003.01	-2.10	-0.05%

5. Methodological & Statistical Interpretation

- **Effect Size Evaluation:** The observed difference in average total interactions between the two cohorts is only 43.97 interactions (Cohen's $d < 0.02$), indicating a statistically and practically negligible effect.
- **Complete-Case Consistency:** When filtering strictly to complete cases where likes are non-null, the gap narrows further to a mere 2.10 interactions (0.05%), proving that missingness in likes does not alter the conclusion.
- **Conclusion:** The SQL analysis decisively refutes the hypothesis that high follower users achieve higher engagement per post. Audience reach and per-post engagement operate as independent dimensions in this dataset.